MAT1575, Classwork15, Fall2026

5.3 The Divergence and the Integral Tests (p.413-422)

1. *Theorem*: **Divergence Test**.

If $\lim_{n\to\infty} a_n = c \neq 0$ or $\lim_{n\to\infty} a_n$ does not exist, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

$a_n \to 0$

$\sum_{n=1}^{\infty} \frac{1}{n}$ diverges

2. If $\lim_{n\to\infty} a_n = 0$, then does $\sum_{n=1}^{\infty} a_n$ diverge? NOT necessarily, inconclusive

3. Show each given series diverges by using Divergence Test. If the test fails, indicate that the it is *inconclusive*.

(a) $\sum_{n=1}^{\infty} \frac{n}{3n-1}$ $a_n = \frac{n}{3n-1}$

Since $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{n}{3n-1} = \frac{1}{3} \neq 0$, then, by divergence test $\sum_{n=1}^{\infty} \frac{n}{3n-1}$ diverges

(b) $\sum_{n=1}^{\infty} \frac{1}{n^3}$ $a_n = \frac{1}{n^3}$ $\lim_{n\to\infty} a_n = 0$ inconclusive

(c) $\sum_{n=1}^{\infty} e^{1/n^2}$ $a_n = e^{\frac{1}{n^2}}$

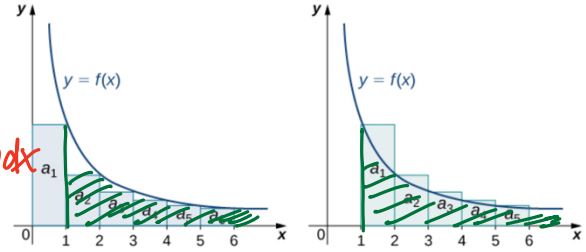
since $\lim_{n\to\infty} a_n = \lim_{n\to\infty} e^{\frac{1}{n^2}} = e^0 = 1 \neq 0$,

by divergence test, $\sum_{n=1}^{\infty} e^{\frac{1}{n^2}}$ diverges

4. *Theorem*: **Integral Test**

Let $\sum_{n=1}^{\infty} a_n$ be a series with $a_n > 0$. Suppose there exists a function f and a positive integer N such that (i) $\boldsymbol{f}$ **is continuous**, (ii) $\boldsymbol{f}$ **is decresing**, and (iii) $\boldsymbol{f(n) = a_n}$ **for all integers** $\boldsymbol{n \geq N}$.

Then $\sum_{n=1}^{\infty} a_n$ and $\int_N^{\infty} f(x)\,dx$ both converge or both diverge.



CAUTION: This Theorem doesn't imply $\sum_{n=1}^{\infty} a_n = \int_1^{\infty} f(x)\,dx$

5. Show the convergence of each given series by using Integral Test.

(a) $\sum_{n=1}^{\infty} \frac{1}{n^3}$ $f(x) = \frac{1}{x^3}$ ① continuous, ② decreasing ③ $f(n) = a_n$

since $\int_1^{\infty} \frac{dx}{x^3}$

(b) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{2n-1}}$

(c) $\sum_{n=1}^{\infty} \frac{1}{n}$

6. *Definition*: p-Series

For any real number p, the series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is called a p series

7. **The convergence of *p*-Series**:

If $p < 0$, then $\frac{1}{n^p} \to$ $as\ n \to \infty$. By , $\sum_{n=1}^{\infty} \frac{1}{n^p}$

If $p = 0$, then $\frac{1}{n^p} \to$ $as\ n \to \infty$. By , $\sum_{n=1}^{\infty} \frac{1}{n^p}$

If $p > 0$, then $f(x) =$ is a , , . Therefore, for $p > 0$, we can use the

- If $p = 1$, $\sum_{n=1}^{\infty} \frac{1}{n}$
- If $p \neq 1$,

In conclusion,

8. Show the convergence of each given series:

$(a) \sum_{n=1}^{\infty} \frac{1}{n^4}$ $(b) \sum_{n=1}^{\infty} \frac{1}{n^{2/3}}$

5.4 Comparison Tests I (p.426-433)

1. *Theorem*: **Comparison Test**

Suppose there exists an integer N such that $0 \leq a_n \leq b_n$ for all $n \geq N$. Then we have

(i) If $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ converges. (ii) If $\sum_{n=1}^{\infty} a_n$ diverges, then $\sum_{n=1}^{\infty} a_n$ diverges.

2. For each given series, use the comparison test to determine whether the series converges or diverges.

$(a) \sum_{n=1}^{\infty} \frac{1}{n^3 + 3n + 1}$

$(b) \sum_{n=1}^{\infty} \frac{1}{2^n + 1}$

$(c) \sum_{n=1}^{\infty} \frac{1}{\ln(n)}$

$(d) \sum_{n=1}^{\infty} \frac{1}{n^2 - 1}$